Assignment 1: Exploration of LLMs

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Introduction

Initially, Large Language Models (LLMs) are an innovation built on top of mathematics, probability, and code snippets. But, given these, the LLMs have integrated and built on neural networks or transformers to gain the ability to exhibit human-like language behaviour and respond to the prompts. So, LLMs have advanced artificial intelligence and natural language processing beyond initial expectations of AI development. They do not just generate the text based on understanding; they operate at a level that sometimes outperforms human benchmarks (Ferraris et al., 2025). At the heart of these models sits the Transformer architecture. These systems use so many parameters, trained on lots of real-world textual datasets. Additionally, they help us in generating text, translation, question answering, and sentiment analysis, setting an innovation for machine language capabilities.

The term “LLM” typically refers to Transformer-based systems with over 10 billion parameters. However, the number of parameters is not their only defining feature. These models demonstrate capabilities inaccessible to smaller architectures. As we can see, GPT-3 handles few-shot prompts and learns from context, a capability which is not in GPT-2.

LLMs have moved beyond their traditional applications and now translate languages, summarise texts, interact with users, and generate content (Jiao et al., 2024). These advances are altering human-computer interaction and transforming how industries process information using right approaches to work with these models.

Yet, they have some limitations. LLMs require really great computational power(GPUs or TPUs), often leading to biases from training data, and occasionally generate false or misleading information, a phenomenon which is known as “hallucinations”. As part of my research, I understand researchers are currently focused on improving efficiency, mitigating ethical risks, and ensuring responsible deployment.

This paper first reviews LLM architecture, then explores applications across industries, discusses challenges and ethics, and concludes with reflections on future impact.

Development and Architecture

This section traces the evolution from traditional models to transformers. Early language models predicted text statistically, but modern LLMs reason through complex problems (Li Y. , 2024). This transition required architectures capable of understanding context across entire passages rather than just neighbouring words. Recurrent and convolutional networks were used before Transformers but had some constraints. RNNs processed tokens one-by-one, making parallelization impossible and causing early context to fade in long sequences. CNNs parallelized computation but could only see fixed local windows (Ferraris et al., 2025). Established that limitations meant neither architecture could truly understand language, where meaning often depends on distant relationships.

Transformers overcame previous limitations by using attention mechanisms. In this system, every token is assigned a query, key, and value vector. The model compares queries against keys to calculate attention weights, which then sum up the values into a rich, context-aware representation. This mechanism is crucial because it allows any word to directly “attend” to any other word, making distant connections in a sentence just as easy to process as immediate neighbours.

The architecture typically involves two main parts. The encoder stacks layers of multi-head self-attention and feed-forward networks, using residual connections to keep training stable. By running “multi-head” attention, the system can process several different types of relationships simultaneously. The decoder works similarly but adds cross-attention to focus on the encoder’s work and uses masking to ensure it only looks at past tokens, enforcing a strict left-to-right generation order (Ferraris et al., 2025). These components can be mixed in order to create Encoder-Decoder/Decoder-Only/Encoder-Only configurations.

What makes modern architecture truly stand out is the emergent behaviour that appears at scale. While scaling proves that performance improves predictably with more data and compute, developers now use efficiency techniques like LoRA and quantization to manage the costs of architecture. It remains surprising that a system designed simply to predict the next token can, when scaled up enough with the right parameters, display capabilities that look like a human mind reasoning.

Applications

LLMs left the computer labs and landed in real-world industries, showing off both their strengths and limits. Healthcare stands out as one of the most promising and cautious areas. These models help pull out important information from medical records, support patient consultations, and simplify complex reports. Evaluations show that ChatGPT generates relatively accurate information across medical disciplines, including genetics, radiation oncology, and biomedicine (Chang et al., 2024). However, there is a risk of fabricated medical advice and limitations in reliably citing sources mean humans have to stay in the loop.

Education has also integrated these a lot. ChatGPT is showing the ability to generate fluent feedback that can do better than human teachers in some contexts, by checking student assignments and providing guidance on task completion (Chang et al., 2024). In educational exams, ChatGPT achieves around 71.8% correctness, comparable to average student scores (Chang et al., 2024). I find it interesting that students with LLM help sometimes actually outperform their peers.

In Finance, LLMs are used for sentiment analysis, claim detection, and reasoning jobs. Domain-specific benchmarks like FLUE evaluate financial LLMs on tasks spanning sentiment analysis to question answering. However, financial applications are particularly sensitive, as errors can lead to significant real-world consequences, so accuracy matters here more than ever to avoid the wrong outputs.

In law, LLMs are impressive at analysing and interpreting documents. GPT-4, for instance, placed in the top 10% on the bar exam (Chang et al., 2024). Across every field, it's more or less the same thing: LLMs look great in limited usage, but real-world integration is tricky since mistakes can have real consequences. The gap between test scores is not dependable; everyday use is still a big no unless we use it with the right guidance. At their best, these models act as smart assistants, boosting human judgment, not replacing. This shows both the promise and risks of relying on LLMs in real-world domains.

Challenges and Ethical Considerations

LLMs raise concerns that go far beyond technical errors- they have real social implications. Hallucination is a major issue. These models can produce answers that sound plausible but are not natural. Sometimes they invent false statistics and dates; other times, they create citations for articles or cases that don't exist at all, including fake citations/fake references/hallucinated people. There have even been incidents where lawyers submitted briefs full of fake case citations, all due to LLMs. This highlights a fundamental tension: prioritizing fluency over factual accuracy, and people often can't tell the difference.

Bias is another problem. LLMs absorb and repeat familiar prejudices-gender, race, class, and more. Research indicates these models assume nurses are women and doctors are men, especially in translation scenarios, just reinforcing old stereotypes (Ferraris et al., 2025; Jiao et al., 2024). What concerns me most is how these biases become subtler as the models advance.

Privacy is also a major vulnerability. If sensitive information ends up in the training data, someone with the right prompts can extract it (Jiao et al., 2024). And when a model outputs something harmful, it's nearly impossible to trace how it happened-the reasoning process is a black box again.

We also tend to overestimate these models. If an answer is long, detailed, and confident, users trust it-even when it's completely inaccurate.

All these issues point to something bigger: we've rushed to adopt LLMs before fully understanding where and how they fail. Solving this isn't just a technical challenge. It requires collaboration among developers, researchers, and open source contributors. Otherwise, we're just fixing problems after the damage is done.

Conclusion

Large language models show us a huge advancement in AI. They are based on the Transformer architecture, which finally addressed many issues with sequential processing and capturing long-range dependencies. With self-attention, these models understand the context between entire passages, not just sentence by sentence. The scaling laws are straightforward: if we provide more data, more computational power, and increase their size, these models steadily improve. There are three primary configurations today: encoder-decoder for tasks like translation and summarisation, decoder-only for text generation, and encoder-only for understanding and classification.

But there is still a risky gap. Some real challenges persist for all this new tech. Hallucinations are a big one. These models sometimes produce entirely incorrect responses, as if they were facts. Privacy is another worry: sensitive information can slip through in unexpected ways. And because these systems are essentially black boxes, when something goes out of hand, it is hard to find who was responsible.

So where does this leave us? LLMs are advancing. They're already changing how we find, process, and interact with information. In future, Models that might be able to say "I don't know" when they genuinely do not know. Transparency in how decisions get made. And always, we need humans in the loop, ready to step in and correct if needed. These aren't just technical puzzles to solve. There are questions about how we, as a society, want to live alongside tools whose full workings we still don't completely grasp.

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